DESIGN AND APPLY CONTEXT-AWARE ACCESS CONTROLS BASED ON USER IDENTITY, DEVICE POSTURE, LOCATION, AND APPLICATION RISK

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INTRODUCTION

- In the modern digital era, organizations depend greatly on cloud platforms, mobile devices, and remote working systems. Due to this shift, traditional security methods that rely only on network boundaries are no longer sufficient. Cybersecurity threats like unauthorized access, data leaks, and system weaknesses are growing quickly, which makes strong security measures necessary to protect important and sensitive data.
- To solve these problems, this project aims to design and develop **context-aware access control mechanisms** that make smart security decisions by analysing different factors such as user identity, device condition, location, and application risk level. Rather than allowing access based only on a username and password, the system examines the overall context of each access request and decides whether the request should be permitted, limited, or denied.
- The project is based on the **Zero Trust security model**, which follows the concept of "never trust, always verify." According to this approach, every user and device must be verified continuously before gaining access to organizational resources. By using contextual data and defined security policies, the system allows only authorized users to access specific applications while blocking unauthorized users and potential cyber threats.

ABSTRACT

- As cloud computing, mobile technologies, and remote working continue to expand, traditional network security approaches are becoming less effective in protecting organizational information. Many organizations are now facing growing cybersecurity risks such as unauthorized access, unsafe devices, and insecure network connections. Because of this, there is a strong need for a more advanced and flexible security system that can evaluate access requests using several factors instead of depending only on usernames and passwords.
- This project aims to design and implement a context-aware access control system that enhances security by examining various contextual factors including user identity, device condition, geographical location, and the risk level of applications. The system analyses these parameters in real time and makes intelligent decisions such as granting access, limiting certain activities, requesting additional authentication, or blocking the request completely when the risk level is high.
- The system is developed based on the Zero Trust security model, where every access request must be verified continuously before permission is granted. By integrating contextual information with predefined security policies, the system helps minimize unauthorized access, defend against cyber threats, and secure sensitive organizational data. The primary

aim of this project is to develop a secure, adaptable, and efficient access control framework that strengthens cybersecurity while ensuring a smooth and convenient user experience.

USE CASE SCENARIOS

Scenario 1:

Employee Access to Company Application

- An employee wants to access the company's internal application from the office using a company-managed laptop. The system checks the employee's identity, device security status, and network location. Since the device is secure and the user is authorized, the system grants full access to the application. This ensures smooth and secure access for trusted users.

Scenario 2: Contractor Access to Project Repository

- A contractor working on a short-term project needs access only to a specific project repository. The system verifies the contractor's identity and device security. Based on predefined policies, the contractor is allowed to access only the project repository during working hours while other company resources remain restricted.

Scenario 3: Executive Access from a Foreign Location

- A company executive tries to access sensitive financial documents while traveling abroad. The system detects an unusual location and identifies the application as high risk. Therefore, the system requires additional authentication such as Multi-Factor Authentication (MFA) and restricts download permissions to protect sensitive information.

Scenario 4: Access from Public Wi-Fi

- An employee attempts to access company data from a coffee shop using public Wi-Fi. Although the user and device are verified, the network is considered risky. The system allows limited access through a secure browser environment to prevent data leakage.

S. No	Scenario / Condition	Result	Description
1	User Role: Admin	Access Granted	System allows admin privileges
2	Device: Laptop	Access Granted	Trusted device detected
3	Location: India	Access Granted	Safe location
4	Application Risk: Low	Access Granted	Low risk application
5	Device: Mobile	Access Denied	Untrusted device
6	Location: Foreign	Access Denied	Suspicious location
7	Risk Evaluation	System Monitoring	Real-time risk checking
8	Device Authentication	Security Context	Device verification process
9	Session Monitoring	Threat Detection	Continuous monitoring
10	Identity Verification	Security Check	User identity validation
12	Application Risk Analysis	Security Control	Risk based access decision

This project DESING AND APPLY CONTEXT-AWARE ACCESS CONTROLS BASED ON USER IDENTITY, DEVICE POSTURE, LOCATION,AND APPLICATION RISK using visual studio code to create a website.

Folder:

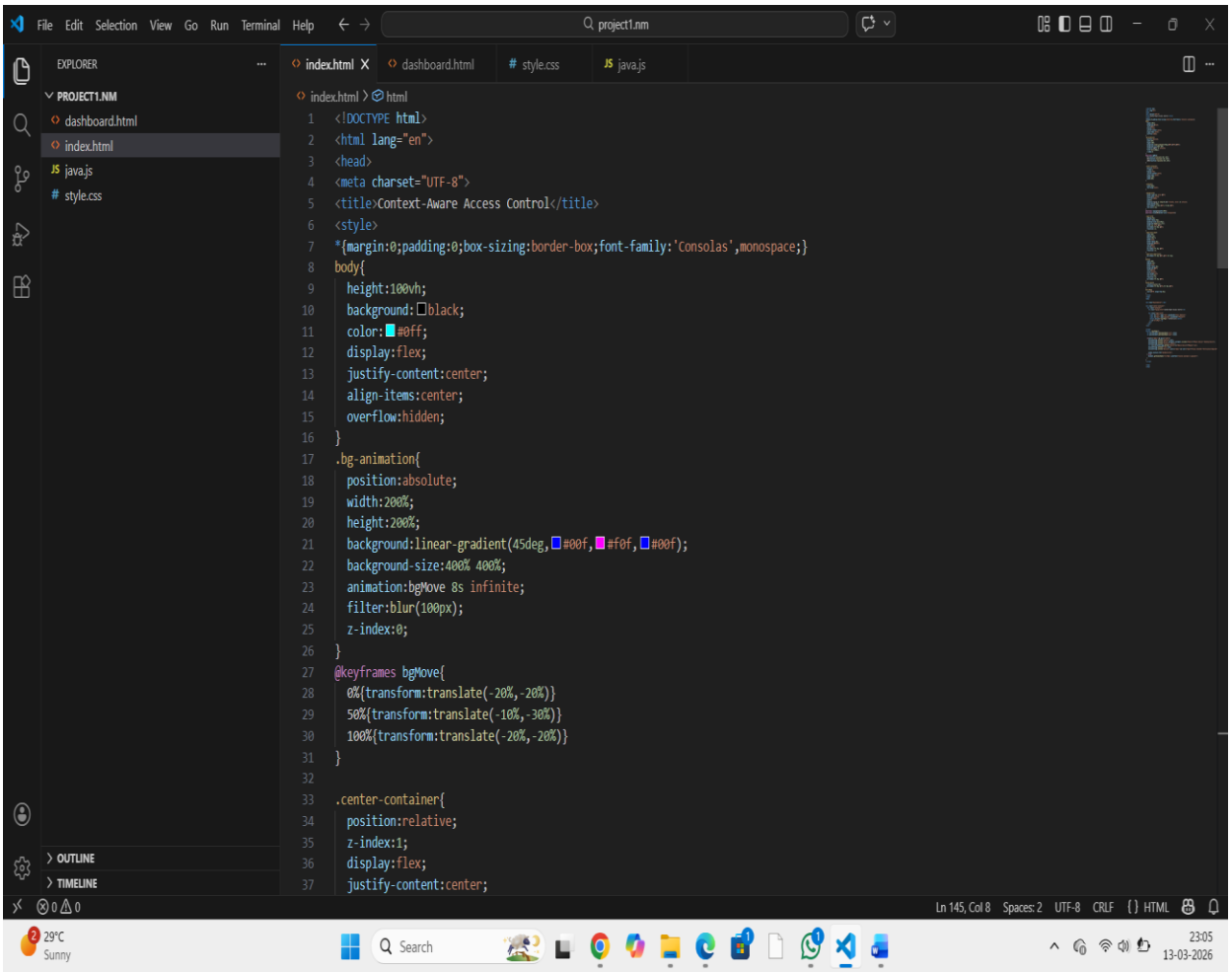
File Name	Type / Language	Purpose
login.html	HTML	Login page where user enters username and password
index.html	HTML	Main **Check Access** page
dashboard.html	HTML	Dashboard page showing system information
about.html	HTML	About the project
logs.html	HTML	Displays access logs or activity
navbar.html	HTML	Navigation bar for the website
access.js	JavaScript	Checks access rules (role, device, location, risk)
login.js	JavaScript	Handles login verification
style.css	CSS	Design and styling for main pages
loginstyle.css	CSS	Styling for login page

CODING SCREENSHORT

File Name: Index.html

Language: HTML

Purpose :Index page where user enters username and password ,Main check access page



```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Context-Aware Access Control</title>
<style>
*{margin:0;padding:0;box-sizing:border-box;font-family:'Consolas',monospace;}
body{
  height:100vh;
  background:black;
  color:#0ff;
  display:flex;
  justify-content:center;
  align-items:center;
  overflow:hidden;
}
.bg-animation{
  position:absolute;
  width:200%;
  height:200%;
  background:linear-gradient(45deg,#00f,#f0f,#00f);
  background-size:400% 400%;
  animation:bgMove 8s infinite;
  filter:blur(100px);
  z-index:0;
}
@keyframes bgMove{
  0%{transform:translate(-20%,-20%)}
  50%{transform:translate(-10%,-30%)}
  100%{transform:translate(-20%,-20%)}
}

.center-container{
  position:relative;
  z-index:1;
  display:flex;
  justify-content:center;
```

```html
<html lang="en">
<head>
<style>
.center-container{
  align-items:center;
  height:100%;
  width:100%;
}

.container{
  width:400px;
  text-align:center;
}

.typing-title{
  border-right:3px solid #0ff;
  white-space:nowrap;
  overflow:hidden;
  width:0;
  animation:typing 3s steps(40,end) forwards, blink .8s infinite;
  margin-bottom:25px;
  text-shadow:0 0 15px #0ff,0 0 25px #0ff;
  font-size:1.5em;
}
@keyframes typing{to{width:100%}}
@keyframes blink{50%{border-color:transparent}}

.login-card{
  padding:25px;
  border-radius:15px;
  backdrop-filter:blur(20px);
  background:rgba(0,0,0,0.6);
  border:1px solid #0ff;
  box-shadow:0 0 40px #0ff;
  transition:.3s;
}
.login-card input{
```

```html
<html lang="en">
<head>
<style>
.login-card input{
  width:100%;
  padding:12px;
  margin:10px 0;
  border:none;
  border-radius:8px;
  background:#020617;
  color:#0ff;
  outline:none;
  box-shadow:0 0 10px #0ff;
  transition:.3s;
}
.login-card input:focus{
  box-shadow:0 0 20px #0ff,#ff0 0 0 10px;
}
button{
  width:100%;
  padding:12px;
  border:none;
  border-radius:8px;
  background:#0ff;
  color:black;
  cursor:pointer;
  font-weight:bold;
  font-size:15px;
  transition:.3s;
  box-shadow:0 0 10px #0ff;
}
button:hover{
  transform:scale(1.05);
  box-shadow:0 0 20px #ff0,0 0 30px #0ff;
}
#errorMsg{
  color:#ff0; margin-top:10px;
```

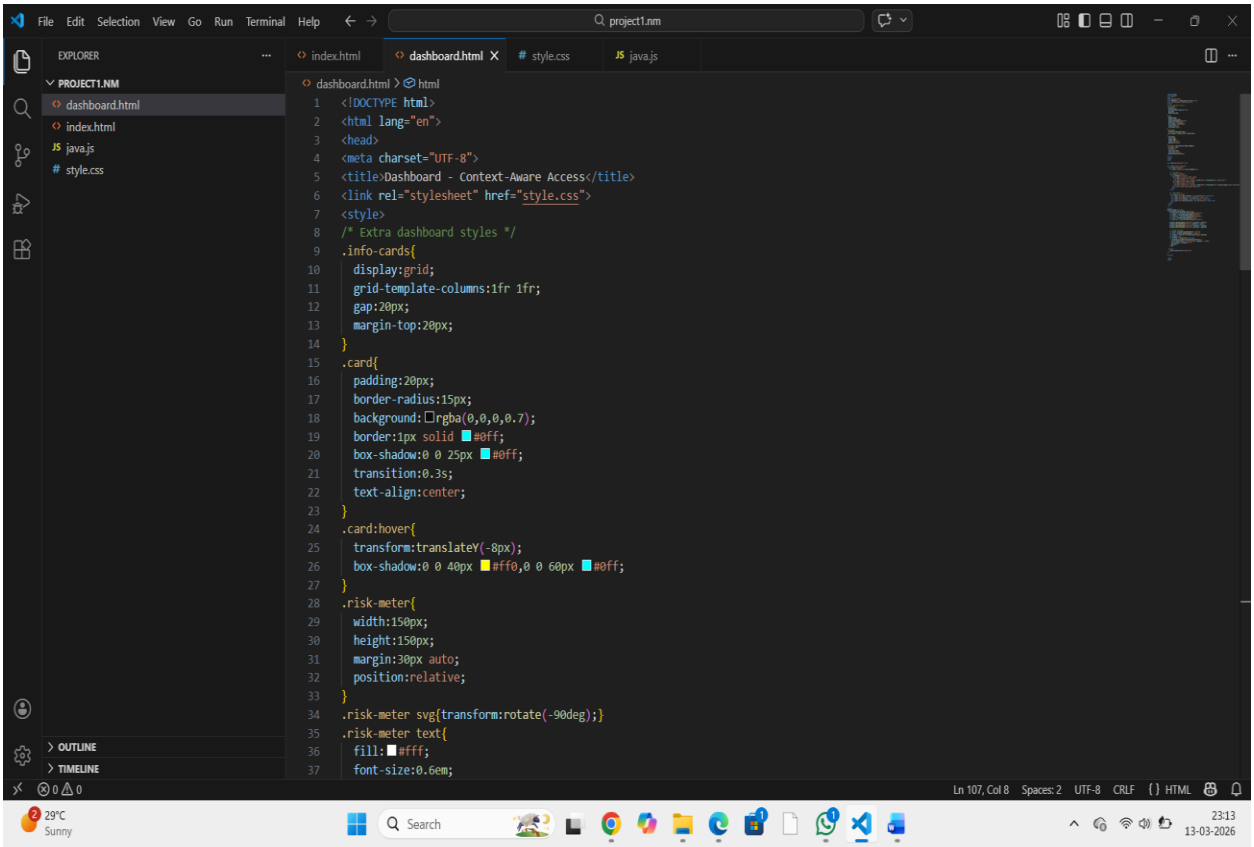
```html
<html lang="en">
<head>
<style>
#errorMsg{
  color: #ff0; margin-top:10px;
}
</style>
</head>
<body>

<div class="bg-animation"></div>

<div class="center-container">
  <div class="container">
    <h1 class="typing-title">Context-Aware Access Control</h1>

    <div class="login-card">
      <input id="user" type="text" placeholder="User Identity">
      <input id="pass" type="password" placeholder="Password">
      <button onclick="loginPage()">Authenticate</button>
      <p id="errorMsg"></p>
    </div>
  </div>
</div>

<script>
function loginPage(){
  let user=document.getElementById("user").value;
  let pass=document.getElementById("pass").value;

  if(user=="admin" && pass=="1234"){
    sessionStorage.setItem("identity",user);
    sessionStorage.setItem("device",navigator.userAgent.includes("Mobile")?"Mobile Device":"Desktop Device");
    sessionStorage.setItem("location","India");
    let risk=(sessionStorage.getItem("device")=="Mobile Device")?"Medium":"Low";
    sessionStorage.setItem("risk",risk);
    sessionStorage.setItem("decision",(user=="admin" && risk!=="High")?"Access Granted":"Verification Required");
```

```html
<html lang="en">
<body>
<div class="center-container">
  <div class="container">
    <div class="login-card">
      <input id="pass" type="password" placeholder="Password">
      <button onclick="loginPage()">Authenticate</button>
      <p id="errorMsg"></p>
    </div>
  </div>
</div>

<script>
function loginPage(){
  let user=document.getElementById("user").value;
  let pass=document.getElementById("pass").value;

  if(user=="admin" && pass=="1234"){
    sessionStorage.setItem("identity",user);
    sessionStorage.setItem("device",navigator.userAgent.includes("Mobile")?"Mobile Device":"Desktop Device");
    sessionStorage.setItem("location","India");
    let risk=(sessionStorage.getItem("device")=="Mobile Device")?"Medium":"Low";
    sessionStorage.setItem("risk",risk);
    sessionStorage.setItem("decision",(user=="admin" && risk!=="High")?"Access Granted":"Verification Required");

    window.location.href="dashboard.html";
  } else{
    document.getElementById("errorMsg").innerText="Invalid username or password!";
  }
}
</script>

</body>
</html>
```

File Name: dashboard.html

Language: HTML

Purpose : Dashboard page showing system information



```html
<html lang="en">
<head>
<style>
.risk-meter text{
  text-anchor:middle;
  dominant-baseline:middle;
}
</style>
</head>
<body>

<div class="bg-animation"></div>

<div class="center-container">
  <div class="container">
    <h1 class="typing-title">User Dashboard</h1>

    <!-- Risk Meter -->
    <div class="risk-meter">
      <svg viewBox="0 0 36 36">
        <path class="circle-bg" d="M18 2.0845
          a 15.9155 15.9155 0 0 1 0 31.831
          a 15.9155 15.9155 0 0 1 0 -31.831" stroke="#111" stroke-width="3.8" fill="none"/>
        <path class="circle" d="M18 2.0845
          a 15.9155 15.9155 0 0 1 0 31.831
          a 15.9155 15.9155 0 0 1 0 -31.831" stroke="#0ff" stroke-width="3.8" stroke-linecap="round" fill="none" stroke-dasharray="0,100"/>
        <text x="18" y="20" id="riskText">0%</text>
      </svg>
    </div>

    <!-- Info Cards -->
    <div class="info-cards">
      <div class="card">User Identity: <span id="identity"></span></div>
      <div class="card">Device: <span id="device"></span></div>
      <div class="card">Location: <span id="location"></span></div>
      <div class="card">Access Decision: <span id="decision"></span></div>
    </div>
```

```html
<html lang="en">
<body>
</div>
<script>
window.onload=function(){
  if(sessionStorage.getItem("identity")){
    let identity = sessionStorage.getItem("identity");
    let device = sessionStorage.getItem("device");
    let location = sessionStorage.getItem("location");
    let risk = sessionStorage.getItem("risk");
    let decision = sessionStorage.getItem("decision");

    document.getElementById("identity").innerText = identity;
    document.getElementById("device").innerText = device;
    document.getElementById("location").innerText = location;
    document.getElementById("decision").innerText = decision;

    // Animate Risk Meter
    let circle = document.querySelector(".circle");
    let text = document.getElementById("riskText");
    let riskVal = (risk=="Low")?30:(risk=="Medium")?60:100;
    let progress = 0;
    let interval = setInterval(function(){
      if(progress>=riskVal){clearInterval(interval);}
      circle.setAttribute("stroke-dasharray", progress + ",100");
      text.innerText = progress + "%";
      progress++;
    },20);

  } else {
    window.location.href="login.html";
  }
}
</script>
</body>
</html>
```

File Name: java.js

Language: JavaScript

Purpose : Checks access rules(role,device,location,risk)

```javascript
window.onload = function(){
  if(sessionStorage.getItem("identity")){
    let identity = sessionStorage.getItem("identity");
    let device = sessionStorage.getItem("device");
    let location = sessionStorage.getItem("location");
    let risk = sessionStorage.getItem("risk");
    let decision = sessionStorage.getItem("decision");

    document.getElementById("identity").innerText = identity;
    document.getElementById("device").innerText = device;
    document.getElementById("location").innerText = location;
    document.getElementById("decision").innerText = decision;

    // Animate Risk Meter
    let circle = document.querySelector(".circle");
    let text = document.getElementById("riskText");

    // Risk % based on Low/Medium/High
    let riskVal = (risk=="Low")?30:(risk=="Medium")?60:100;

    let radius = 15.9155; // as per SVG circle
    let circumference = 2 * Math.PI * radius;

    circle.setAttribute("stroke-dasharray", `0 ${circumference}`); // start from 0
    let progress = 0;

    let interval = setInterval(function(){
      if(progress > riskVal){
        clearInterval(interval);
      } else {
        let dash = (progress/100)*circumference;
        circle.setAttribute("stroke-dasharray", `${dash} ${circumference}`);
        text.innerText = progress + "%";
        progress++;
      }
    },20);

```

File Edit Selection View Go Run Terminal Help

project1.nm

EXPLORER

PROJECT1.NM

- dashboard.html
- index.html
- java.js
- style.css

index.html | dashboard.html | style.css | java.js

java.js > onload

```javascript
window.onload = function(){
    document.getElementById("device").innerText = device;
    document.getElementById("location").innerText = location;
    document.getElementById("decision").innerText = decision;

    // Animate Risk Meter
    let circle = document.querySelector(".circle");
    let text = document.getElementById("riskText");

    // Risk % based on Low/Medium/High
    let riskVal = (risk=="Low")?30:(risk=="Medium")?60:100;

    let radius = 15.9155; // as per SVG circle
    let circumference = 2 * Math.PI * radius;

    circle.setAttribute("stroke-dasharray", `0 ${circumference}`); // start from 0
    let progress = 0;

    let interval = setInterval(function(){
      if(progress > riskVal){
        clearInterval(interval);
      } else {
        let dash = (progress/100)*circumference;
        circle.setAttribute("stroke-dasharray", `${dash} ${circumference}`);
        text.innerText = progress + "%";
        progress++;
      }
    },20);

  } else {
    window.location.href="login.html";
  }
}
```

OUTLINE

TIMELINE

0 0

Ln 41, Col 2 Spaces: 2 UTF-8 CRLF {} JavaScript

29°C Sunny

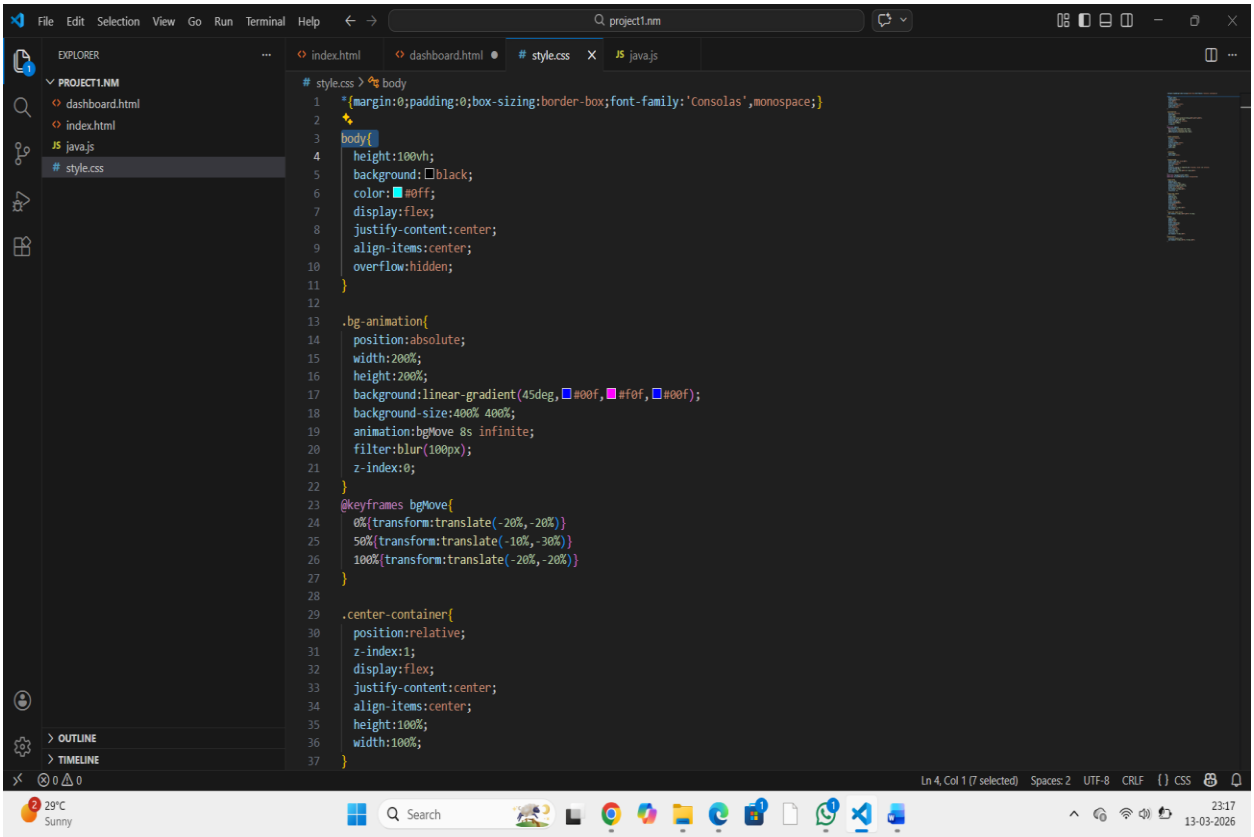
Search

23:21 13-03-2026

File Name: style.css

Language: CSS

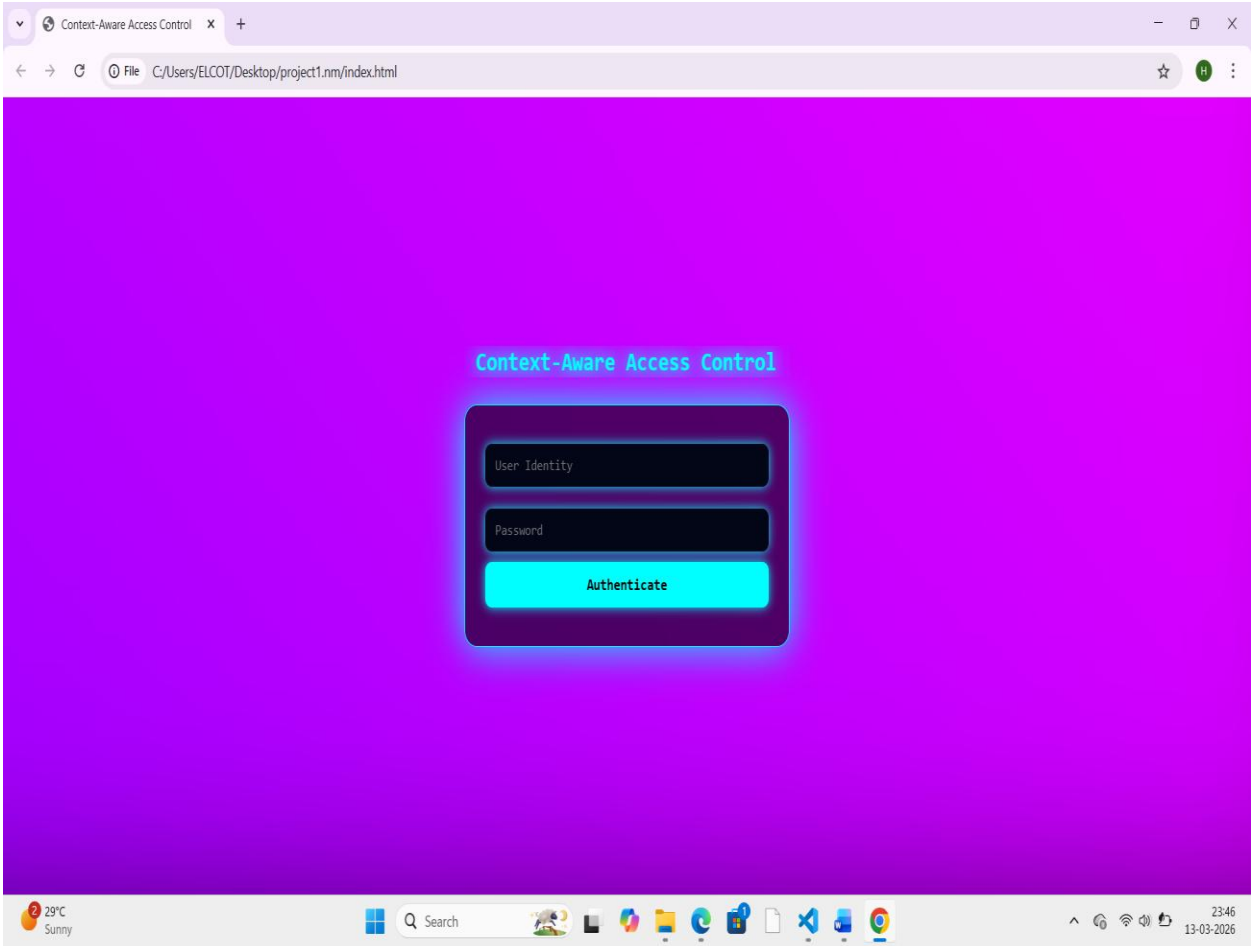
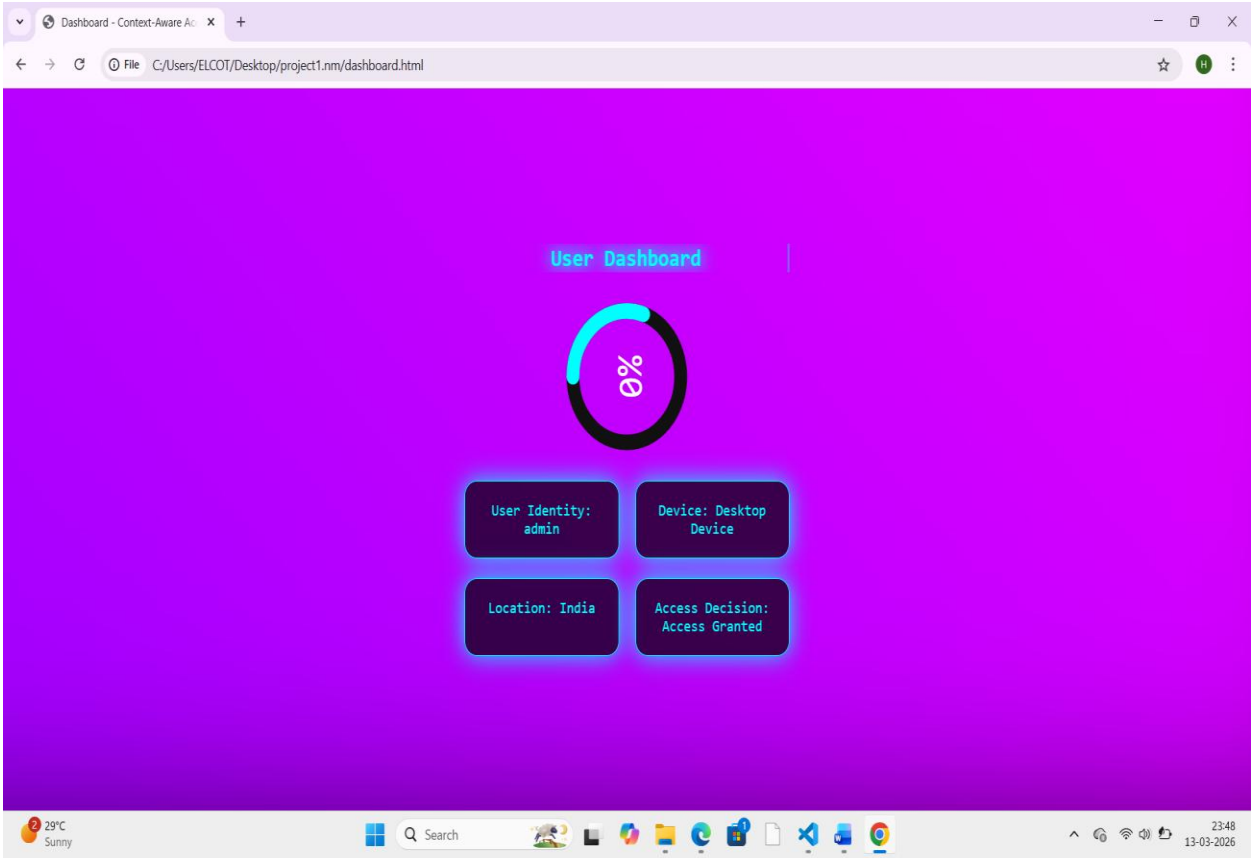
Purpose : Design and styling for main pages



```css
.container{
  width:400px;
  text-align:center;
}

.typing-title{
  border-right:3px solid #0ff;
  white-space:nowrap;
  overflow:hidden;
  width:0;
  animation:typing 3s steps(40,end) forwards, blink .8s infinite;
  margin-bottom:25px;
  text-shadow:0 0 15px #0ff,0 0 25px #0ff;
  font-size:1.5em;
}
@keyframes typing{to{width:100%}}
@keyframes blink{50%{border-color:transparent}}

.login-card{
  padding:25px;
  border-radius:15px;
  backdrop-filter:blur(20px);
  background:rgba(0,0,0,0.6);
  border:1px solid #0ff;
  box-shadow:0 0 40px #0ff;
  transition:.3s;
}
.login-card input{
  width:100%;
  padding:12px;
  margin:10px 0;
  border:none;
  border-radius:8px;
  background:#020617;
  color:#0ff;
  outline:none;
```

```css
.login-card{
  border:1px solid #0ff;
  box-shadow:0 0 40px #0ff;
  transition:.3s;
}
.login-card input{
  width:100%;
  padding:12px;
  margin:10px 0;
  border:none;
  border-radius:8px;
  background:#020617;
  color:#0ff;
  outline:none;
  box-shadow:0 0 10px #0ff;
  transition:.3s;
}
.login-card input:focus{
  box-shadow:0 0 20px #0ff,#ff0 0 0 10px;
}
button{
  width:100%;
  padding:12px;
  border:none;
  border-radius:8px;
  background:#0ff;
  color:black;
  cursor:pointer;
  font-weight:bold;
  font-size:15px;
  transition:.3s;
  box-shadow:0 0 10px #0ff;
}
button:hover{
  transform:scale(1.05);
  box-shadow:0 0 20px #ff0,0 0 30px #0ff;
}
```

OUTPUT SCREENSHORT:



Context -Aware Access Control Dashboard: